

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem Statement #33 – Micro-Lending & Peer Credit Risk Assessor

Problem context: A peer-to-peer micro-finance platform calculates borrower risk scores using heuristic rules, generates structured repayment schedules, and triggers automated default warnings. The target stakeholders/actors are Borrower and Peer Lender.

1. Requirements Table

1.1 Functional Requirements

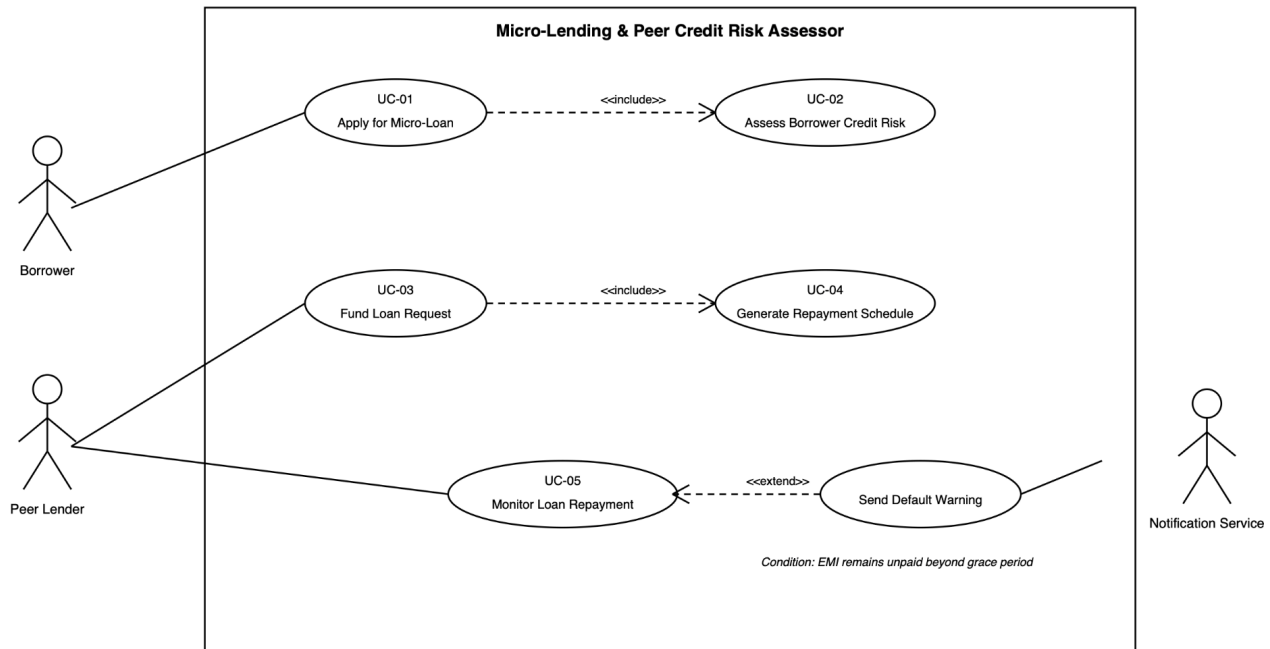
ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute borrower credit risk ratings (Low/Medium/High) based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Pass: Risk tier is accurately assigned for a complete financial profile. Fail: An incomplete financial profile does not receive credit approval.	Core value proposition; downstream lending decisions depend on the risk tier.
FR-002	Functional	The system shall allow a Borrower to submit a micro-loan application specifying loan amount, purpose, income, existing debt, and employment status.	High	Pass: A valid application is stored and enters risk assessment. Fail: An application with missing mandatory fields is rejected.	Entry point of the lending workflow and prerequisite for risk assessment.
FR-003	Functional	The system shall generate a structured EMI repayment schedule, including due dates and principal/interest split, once a loan request is funded, based on the agreed interest rate and tenure.	High	Pass: The schedule covers the agreed tenure and total scheduled repayment equals principal plus interest within rounding tolerance. Fail: The generated schedule has incorrect totals or dates.	Required by the problem statement and needed to track borrower obligations and lender returns.
FR-004	Functional	The system shall allow a Peer Lender to browse open loan requests filtered by risk tier and fund a selected request partially or fully.	Medium	Pass: The funded amount is deducted from the lender's available balance and the loan status is updated. Fail: The lender cannot fund an amount exceeding available balance.	Enables the peer-to-peer lending mechanism.
FR-005	Functional	The system shall automatically trigger a default warning to the Borrower and Peer Lender when an EMI remains unpaid beyond a configurable grace period after its due date.	High	Pass: A warning is generated within 24 hours after grace-period expiry. Fail: An overdue EMI has no warning after the grace period.	Required by the problem statement and protects lender capital while alerting the borrower.

1.2 Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	Loan repayment transactions and EMI distribution shall execute inside database ACID transactions to prevent double-spending.	High	Pass: Concurrency tests show no duplicate or lost EMI postings and transactions meet the target latency under simulated peak load. Fail: Double-spending, lost postings, or unacceptable transaction latency occurs.	Financial correctness under concurrent transactions is essential.

NFR-002	Security & Privacy	Sensitive borrower financial data (income, debt, and repayment history) shall be encrypted at rest and in transit, with role-based access control restricting access to authorized fields.	High	Pass: Security/audit tests verify encryption and access control; a Peer Lender cannot retrieve raw income/debt fields. Fail: Unauthorized financial data is accessible.	Protects sensitive financial information and prevents misuse.
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2. UML Use-Case Model



Actor	Role
Borrower	Submits loan applications and makes EMI repayments.
Peer Lender	Browses loan requests, funds loans, and monitors repayments.
Notification Service	External supporting system used to deliver automated default warnings.

Primary use cases:

- Apply for Micro-Loan
- Assess Borrower Credit Risk
- Fund Loan Request
- Generate Repayment Schedule
- Make EMI Repayment
- Monitor Loan Repayment
- Send Default Warning

3. Use-Case Flow Specification

Field	Detail
Use Case ID	UC-01
Use Case Name	Apply for Micro-Loan
Primary Actor	Borrower
Includes	UC-02: Assess Borrower Credit Risk

3.1 Preconditions

1. Borrower is registered and authenticated on the platform.
2. Borrower does not have another pending loan application that conflicts with the platform's lending rules.

3.2 Postconditions

Success: The loan application is stored with an assigned risk tier and is made available to Peer Lenders for funding.

Failure: No invalid application is approved or published; the Borrower is informed of the missing or invalid information.

3.3 Main Success Scenario

3. Borrower selects “Apply for Micro-Loan” from the dashboard.
4. The system displays an application form requesting loan amount, purpose, income, existing debt, and employment status.
5. Borrower enters the required information and submits the application.
6. The system validates that all mandatory fields are present and correctly formatted.
7. System invokes the included use case “Assess Borrower Credit Risk”.
8. The system calculates the debt-to-income ratio and evaluates repayment history and employment stability.
9. System assigns a Low, Medium, or High risk tier to the application.
10. The system stores the application and publishes the loan request to the Peer Lender marketplace with the assigned risk tier.
11. The system confirms successful submission to the Borrower.

3.4 Alternate Flow A1 – Incomplete Financial Profile

Trigger: At Step 4, the system detects that one or more mandatory financial fields are missing or invalid.

12. The system stops processing and does not invoke the credit-risk assessment.
13. The system displays the specific missing or invalid fields to the Borrower.
14. Borrower corrects the information and resubmits the application.
15. Flow resumes at Main Success Scenario Step 4.